

Data

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- Satellite data file
1. tropomi_gosat_pr_xch4_2x2latlon_dailymeans_14_14_6575_7304_20182019_correction_albedo_only.nc: global TROPOMI and GOSAT proxy XCH4 daily collocated means averaged into a 2 x 2 degree grid for the period 1 January 2018–31 December 2019
 2. tropomi_xch4_14_14_corr_only_alb_2yr.nc: global TROPOMI XCH4 data for 2019 averaged in a cylindrical equal-area grid with 0.3 x 0.5 degree resolution at the Equator.

Data processing

Meteorological data-National Centers for Environmental Prediction (NCEP)

- Global forecast system (GFS):
 - atmospheric and land-soil variables, including temperatures, winds, precipitation, soil moisture, and atmospheric ozone concentration
 - The system couples four separate models (atmosphere, ocean model, land/soil model, and sea ice)
- North American Mesoscale Forecast System (NAM):
 - weather parameters, including temperature, precipitation, lightning, and turbulent kinetic energy
 - uses additional numerical weather models to generate high-resolution forecasts over fixed regions
- Rapid Update Cycle (RUC)
 - Numerical forecast model
 - Analysis/assimilation system to initialize the model
- NCEP/NCAR Reanalysis Project (NNRP) <https://rda.ucar.edu/datasets/ds090.0/>
 - 80 different variables, (including geopotential height, temperature, relative humidity, u- and v- wind components, etc.) in several different coordinate systems, such as 17 pressure level stack on 2.5 by 2.5 degree grids, 28 sigma level stack on 192 by 94 Gaussian grids, and 11 isentropic level stack on 2.5 by 2.5 degree grid.

TROPOMI data:

/mnt/d/Project1/data/s5p/ftp.sron.nl/open-access-data-2/TROPOMI/tropomi/ch4/14_14_Lorente_et_al_2020_AMTD

Note: (SAME WITH [35]) CH4 Version 14_14 : Lorente et al. (2021), AMT. Independent bias correction, updated spectroscopy, DEM and regularisation. <https://amt.copernicus.org/articles/14/665/2021/>

Ncview *.nc

- Instrument
- Meteo
- Target_product
- Side_product
- Diagnostics

Data generation explanation: <https://emissions-api.org/data.html>

Data:

- Geographical Static Data:
https://www2.mmm.ucar.edu/wrf/users/download/get_sources_wps_geog.html

- Meteorological fields

<https://rda.ucar.edu/datasets/ds083.2/> D:\Project1\data\NCEP_83

<https://rda.ucar.edu/datasets/ds094.0/> D:\Project1\data\NCEP_FLEXPART_model (NCEP_rename.sh+download_ds094.0.py)

- Initial and 6-hourly boundary condition-CAMS

<https://www.ecmwf.int/en/newsletter/158/meteorology/new-cams-global-reanalysis-atmospheric-composition>

- [39] oil/gas/coal emission <https://doi.org/10.7910/DVN/HH4EUM>
- Anthropogenic emission: EDGARv5
https://edgar.jrc.ec.europa.eu/dataset_ghg50
<https://zenodo.org/record/6130621>
<https://edgar.jrc.ec.europa.eu/gallery?release=v70ghg&substance=CH4§or=TOTALS> (EDGARv7)
- Wetland emission: WeCHARTs version 1.2.1
https://daac.ornl.gov/cgi-bin/dsviewer.pl?ds_id=1502

EDGARv7

1A1a: Power industry

1A1b+1A1c+1A5b1+1B1b+1B2a5+1B2a6+1B2b5+2C1b: Oil refineries and Transformation industry

1A2: Combustion for manufacturing

1A3a_CDS: Aviation climbing&descent

1A3a_CRS: Aviation cruise

1A3a_LTO: Aviation landing&takeoff

1A3a_SPS: Aviation supersonic

1A3b_noRES: Road transportation no resuspension

1A3c+1A3e: Railways, pipelines, off-road transport

1A3d+1C2: Shipping

1A4: Energy for buildings

1B1a+1B2a1+1B2a2+1B2a3+1B2a4+1B2c: Fuel exploitation

2A: Non-metallic minerals production

2B: Chemical processes

2C1a+2C1c+2C1d+2C1e+2C1f+2C2: Iron and steel production

2C3+2C4+2C5: Non-ferrous metals production

2G: Non energy use of fuels

4A: Enteric fermentation

4B: Manure management

4F: Agricultural waste burning

4C+4D: Agricultural soils

6A+6D: Solid waste landfills

6C: Solid waste incineration

6B: Waste water handling

7A: Fossil Fuel Fires